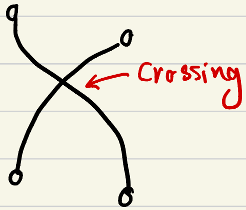
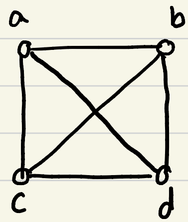
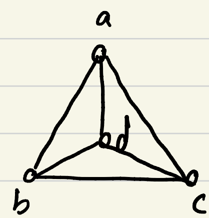
Planar Graphs : a graph is planar/plane if it can be drawn s.t. no two edges cross/interset each other.

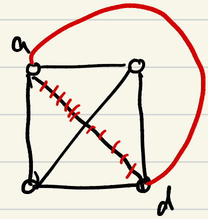




K_4

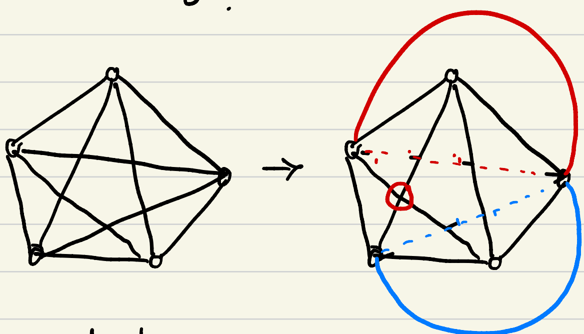


K_4

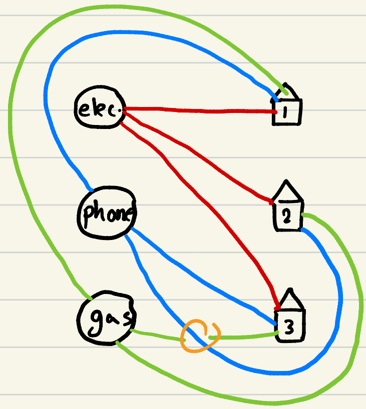


↳ this graph is planar

How about K_5?

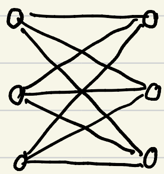


not planar



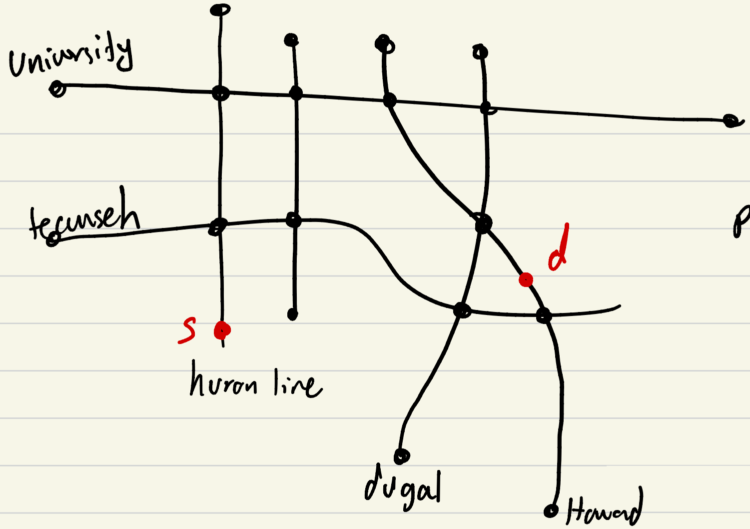
Theorem (Kuratowski) : a graph is planar if and only if it does not have a subgraph of the form K_5 or $K_{3,3}$.

(as a minor)



$K_{3,3}$

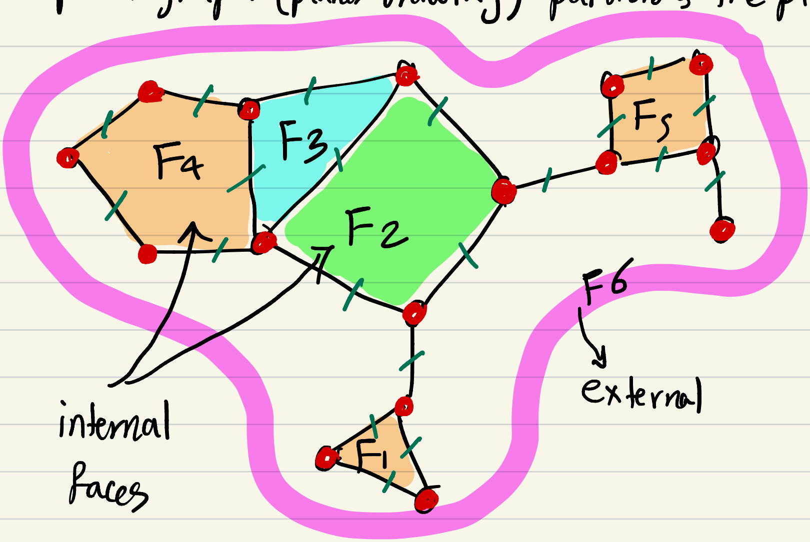
not planar



Planar graph

Euler Formula (for connected planar graphs)

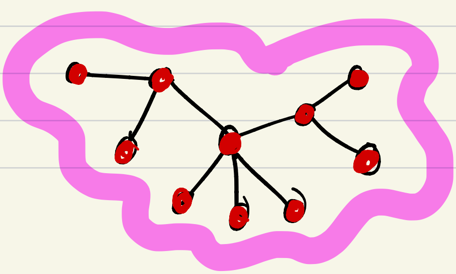
a planar graph (planar drawing) partitions the plane into regions (faces)



Euler Formula

$$\boxed{|V| - |E| + |F| = 2}$$

$$16 - 20 + 6 = 2$$



$$10 - 9 + 1 = 2$$

Proof of Euler Formula:

we prove something stronger

G: planar

$|V|$: # vertices

$|E|$: # edges

$|F|$: # faces

ω : # components

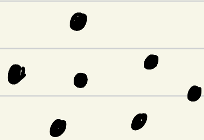
we prove $\boxed{\underbrace{|V|-|E|+|F|}_{LHS} = \underbrace{\omega+1}_{RHS}}$

if G is connected then $\omega = 1$

$\rightarrow |V|-|E|+|F| = 2$ (Euler Formula)

Proof by induction on $|E|$ (keep all vertices and add edges one by one)

- Base Case ($|E|=0$)



$|V|$ vertices

$|E| = 0$

$|F| = 1$

$\omega = |V|$

$LHS = |V|-|E|+|F|$
$= |V| - 0 + 1$
$= |V|+1$

$RHS = \omega + 1 = |V|+1$

LHS = RHS ✓

- I.H.: assume the formula is true for any graph with $|V|$ vertices and $|E'|$ edges where $|E'| < |E|$.

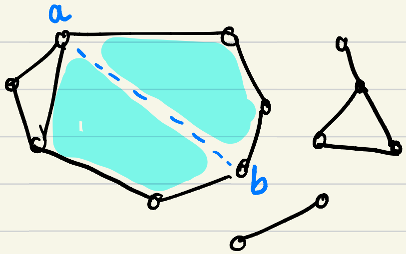
- I.S.: prove that the formula is true for graph G with $|E|$ edges.

Remove an edge ab from the graph. Let E' be the new set of edges in the new graph G'.

$|E'| = |E| - 1$

$|V'|$ $|F'|$ ω'

Case 1: a and b are in same component in G'



G'

$|V'| = |V|$

$|E'| = |E| - 1$

$\omega' = \omega$

$|F'| = |F| - 1$

Use I.H. on G'

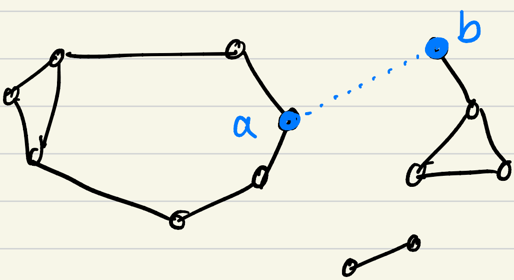
$$|V'| - |E'| + |F'| = \omega' + 1$$

$$|V| - (|E| - 1) + (|F| - 1) = \omega + 1$$

$$|V| - |E| \cancel{+1} + |F| \cancel{-1} = \omega + 1$$

$$|V| - |E| + |F| = \omega + 1 \ \checkmark$$

Case 2: a and b are in different components in G'



G'

$|V'| = |V|$

$|E'| = |E| - 1$

$|F'| = |F|$

$\omega' = \omega + 1$

Use I.H. on G'

$$|V'| - |E'| + |F'| = \omega' + 1$$

$$|V| - (|E| - 1) + |F| = (\omega + 1) + 1$$

$$|V| - |E| \cancel{+1} + |F| = \omega \cancel{+1} + 1$$

$$|V| - |E| + |F| = \omega + 1 \ \checkmark$$

$$|V| - |E| + |F| = \omega + 1$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$v \quad e \quad f$$

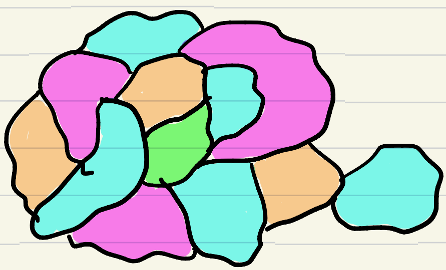
$$\left.\begin{matrix} v - e + f = \omega + 1 \\ \omega \geq 1 \end{matrix}\right\} v - e + f \geq 2 \Rightarrow \boxed{e \leq v + f - 2}$$

$(*)$

<u>Theorem</u>: In any planar graph $\boxed{e \leq 3v - 6}$ and $\boxed{f \leq 2v - 4}$

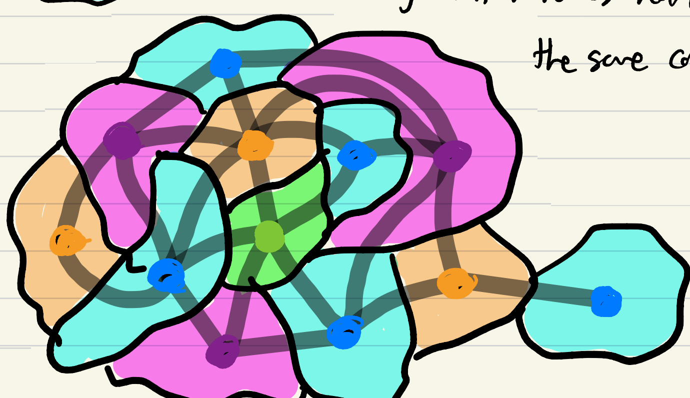
we prove this next week

Coloring of map



graph coloring:

Color vertices with min # of colors s.t no two adjacent vertices have the same color.



Four Color Thm:

any planar graph w/ map can be colored by 4 colors